Hall Ticket Number:

MR-21

Code: B413IT5

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Regular Examinations, Aug/Sep-2023

Discrete Mathematics

(Information Technology)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

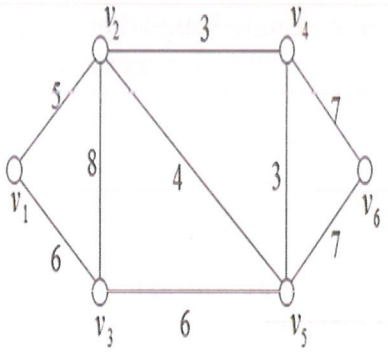
Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Write the converse and contra positive of the statement" If X is odd then X^2 is odd."	2	1	1	1
1. b)	Explain about Quantifiers.	2	1	1	1
1. c)	Explain the properties of relations.	2	1	2	1
1. d)	Define Equivalence relation.	2	1	2	1
1. e)	Develop a recursive algorithm to find factorial of a number.	2	3	3	1
1. f)	Find the number of 5-letter words which contain three different consonants and two different vowels of the English alphabet.	2	3	3	1,2
1. g)	Solve the following recurrence relation by substitution method. $a_n=a_{n-1}+n$ where $a_0=2$.	2	3	4	1,2
1. h)	Write about expected value and variance.	2	1	4	1
1. i)	Define subgraphs with suitable examples.	2	1	5	1
1. j)	Explain spanning trees.	2	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Show that $(\forall x)\{P(x) \vee Q(x)\} \Rightarrow (\forall x) P(x) \vee (\exists x) Q(x)$.	5	2	1	1,2
2. b)	Explain in detail about logical connectives with examples.	5	2	1	1
	OR				
2. c)	Show that $R \rightarrow S$ can be derived from the premises $P \rightarrow (Q \rightarrow S)$, $\neg R \vee P$ and Q.	5	2	1	1,2
2. d)	Construct the truth table for the following statement : $(P \rightarrow Q) \vee (\neg P \rightarrow R)$	5	1	1	1
3. a)	Given the relation matrix M_R of a relation R, $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ Find the matrices of $R^2 = R \circ R$, $R^3 = R \circ R \circ R$.	5	1	2	1
3. b)	Let A= {1, 2, 3, 4, 6, 8, 12, 24}.Show that the relation 'divides' is partial ordering on A. Draw Hasse diagram.	5	2	2	1,2
	OR				
3. c)	LetX={1,2,3,4,5,6}and R={(1,4),(1,3),(1,2),(2,5),(2,4),(2,3),(3,4),(3,6),(5,6)}. Define a compatibility relation. Draw the graph and find the maximal compatibility blocks for the relation R.	5	2	2	1,2
3. d)	What is composition of functions? Let f, g, h be functions from R to R defined by $f(x) = x^2$ and $g(x) = x+5$ and $h(x) = 3x$, for all $x \in R$, Find gof, fog, fof, gog, hof, goh?	5	2	2	1,2
4. a)	Using the principle of mathematical induction, prove that $1 \cdot 3 + 3 \cdot 5 + 5 \cdot 7 + \ldots + (2n - 1)(2n + 1) = (1/3)\{n(4n^2 + 6n - 1)$.	5	3	3	1,2
4. b)	Find the coefficient of $x^3y^3z^2$ in $(2x-3y+5z)^8$.	5	1	3	1,2
	OR				
4. c)	State and prove Binomial theorem.	5	1	3	1

P.T.O.

4. d)	In how many ways can 6 men and 6 women be seated in a row? i) If any person may sit next to any other. ii) If men and women must occupy alternative seats.	5	3	3	1
5. a)	Solve $a_n-7a_{n-1}+10a_{n-2}=0$, $n>=2$, given $a_0=10$, $a_1=41$ using generating functions.	5	3	4	1,2
5. b)	Solve the recurrence relation using substitution method. $a_n= a_{n-1}+3n^2+3n+1$ where $a_0=1$.	5	3	4	1,2
	OR				
5. c)	Solve the recurrence relation $a_n - 4a_{n-1} + 4 a_{n-2} = (n^2+1)2^n$ for $n \geq 2$, given $a_0=0$, $a_1=1$ by generating functions method.	5	3	4	1,2
5. d)	How many different words can be formed with the letters of the word 'SUPER' such that the vowels always come together?	5	3	4	1
6. a)	What is the significance of Planar graphs? Is $K_{3,3}$, planar?	5	1	5	1
6. b)	Differentiate between Eulerian graph and Hamiltonian graph with an example each.	5	4	5	1
	OR				
6. c)	Explain kruskal's algorithm and using the same obtain the minimal spanning tree for the following weighted graph. 	5	3	5	1,2
6. d)	Write algorithms for various tree traversals.	5	1	5	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome